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ARTICLE

Synthesis of 1-Benzylisoindoline and 1-Benzyl Tetrahydroisoquinoline through Nucleophilic Addition of Organozinc Reagents to *N,O*-acetals

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A new approach to access 1-benzylisoindoline and 1-benzyl-tetrahydroisoquinoline has been developed through nucleophilic addition of organozinc reagents to *N,O*-acetals. A number of substituted organozinc reagents were amenable to this transformation, and the desired products were obtained with excellent yields. Moreover, $Sc(OTf)_3$ proved to be an effective catalyst for the formation of 1-benzylisoindoline and 1-benzyl-tetrahydroisoquinoline using such nucleophilic addition.

Introduction

Benzylisoindoline and benzylisoquinoline scaffolds are widely served as key intermediates in the synthesis of indole and isoquinoline alkaloids,[1] which are of great pharmaceutical significance.[2] For instance, laudanosine (**1**), isolated from the roots of *Polyalthia cerasoides*,[3] and lennoxamine (**2**), isolated from barberries species,[4] exhibit potent anticancer, antineuropsychiatric, vasorelaxing, GABA receptor and

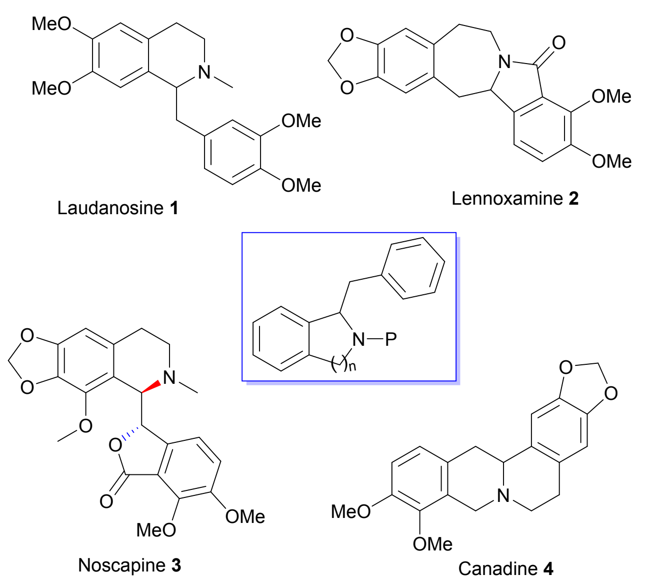

Figure 1. Structures of several natural products.

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anticough activities[5]. Noscapine (**3**), isolated from *Papaver somniferum L.*[6] is an antitussive agent[7a] and displays other potential clinical utilities[7] for the treatment of stroke, anxiety and cancer. Canadine (**4**), a protoberberine alkaloid isolated from *Corydalis yanhusuo*, can act as a calcium channel blocker.[8] (**Figure 1**).

From the synthetic point of view, 1-substituted isoindoline was readily accessible through intramolecular hydroamination of aminoalkenes,[9] Meyers' methodology (from 2, 3-dihydro-1H-isoindole via metallation and alkylation of its *tert*-butyl formamidino derivative),[10] and addition-reduction elimination-reduction of isoindoline-1,3-dione.[11] Among them, only the first method reported the synthesis of 1-benzylisoindoline. Despite these robust processes, a straightforward and efficient approach to 1-benzylisoindoline is still in demand. As for 1-substituted-tetrahydroisoquinoline, recent chemistry efforts led to a number of powerful approaches including haloamide cyclization,[12] Pictet – Spengler reaction,[13] alkylation of anodically prepared α-amino nitriles,[14] de novo BIA cascades,[15] Ir-catalyzed hydrogenation of 1-substituted isoquinolines,[16] and metal and oxidant free C(sp3)–H functionalization.[17] However, some of these methods suffer from harsh reaction conditions or narrow adaptability, the straightforward and efficient approach to 1-benzylisoquinoline is still quite limited. Therefore, the preparation of 1-benzylisoindoline and 1-benzylisoquinoline is still widely pursued, aiming for effective approaches under mild conditions.

In the past decade, slightly functionalized organozinc reagents have been demonstrated many advantages in modern organic synthesis.[18] In addition to their applications in the synthesis of natural products, clinical candidates and drugs,[19] direct nucleophilic addition of organozinc reagents were successfully realized to functional groups like carbonyl,[20] nitrones[21] and imine.[22] Recently, synthetic chemists also focused on their reactions with *N,O*-acetals.[23] Our group have successfully established the approach to 1,4 and 1,5-amino alcohols from

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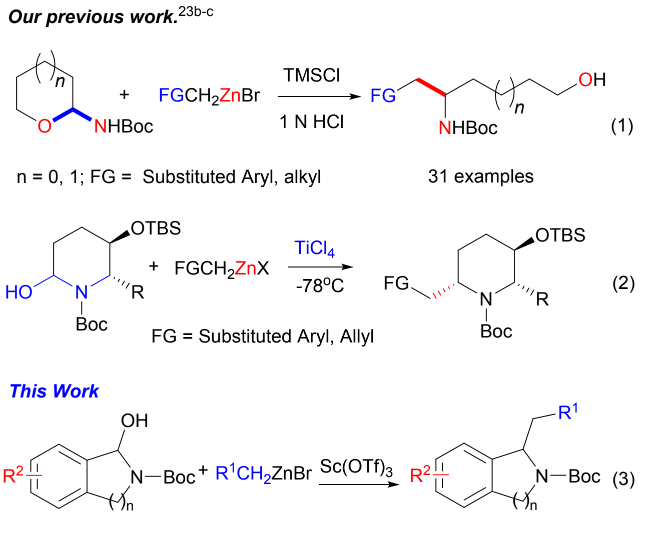
semicyclic *N,O*-acetals through TMSCl-catalyzed nucleophilic addition with functionalized organozinc reagents (Fig 2, eq 1).[23b] Subsequently, we developed a stereoselective method to 2,6-substituted piperidine skeleton through $TiCl_4$-catalyzed nucleophilic substitution (Fig 2, eq 2).[23c] Herein, we present a practical method to functionalized 1-benzylisoindoline and 1-benzylisoquinoline skeletons from *N,O*-acetals **5/6** through $Sc(OTf)_3$ catalyzed nucleophilic substitution process with functionalized organozinc reagents (Fig 2, eq 3).

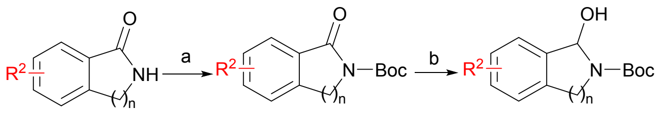


5 (n=1) **6** (n=2) **7** **8** (n=1) **9** (n=2)

Figure 2. Our strategy to access 1-benzylisoindoline and 1-benzyl-tetrahydroisoquinoline scaffolds.

Results and discussion

The *N,O*-acetal substrates **5/6** were readily prepared from lactams **10/11** through N-Boc protection (Boc_2O) and subsequent reduction (DIBAL-H) (Scheme 1).[21] It is worth mentioning that the reduced products (*N,O*-acetal) were not stable and need to be used as soon as possible.

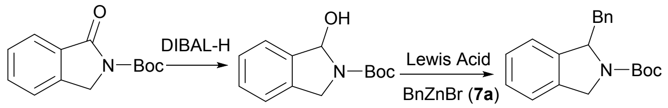


10 (n=1) **11** (n=2) **12** (n=1) **13** (n=2) **5** (n=1) **6** (n=2)

Scheme 1. The synthesis of *N,O*-acetals 5/6. Reagents and conditions: a. Boc_2O, DMAP, THF; b. DIBAL-H, THF, -78 °C.

Our investigation started with the reaction of reduction **12a** to obtain *N,O*-acetal **5a**. Subsequently, the reaction of *N,O*-acetal **5a** with benzylzinc bromide was explored. Initially, the similar lewis acid (TMSCl)[23b] which was used in our previous process was investigated in the nucleophilic substitution of *N,O*-acetal **5a** with benzylzinc bromide, while the yield of the desired **8a** was very low (Tabele 1 entry 1). The yield of **8a** was not improved obviously after several times of exploration. Inexplicably, $TiCl_4$[23c,27], an excellent Lewis acid to catalyzed the reaction of *N,O*-acetal with functionalized alkyl zinc reagents, could not promote this nucleophilic substitution process (Table 1, entry 2). It is worth mentioning that when the mixture of **5a** and benzylzinc bromide was stirred at -78 °C to room temperature for overnight, no desired product **8a** was also observed (Table 1, entry 3). To improve the yield of this nucleophilic process, a variety of Lewis acids were then screened. When $BF_3{\cdot}OEt_2$ and TMSOTf were examined, the yield of **8a** was not dramatically increased (Table 1, entries 4-5). However, both $ZnCl_2$ and $Cu(OTf)_2$ led to the desired **8a**, albeit with low yields (Table 1, entries 6-7). More metal triflates were subsequently tested. $Zn(OTf)_2$ and $In(OTf)_2$ turned out to be ineffective, and no desired product **8a** was observed (Table 1, entries 8-9). To our delight, when $Sc(OTf)_3$ was evaluated, the yield of **8a** was significantly improved to 88% (Table 1, entry 10). Then the equivalent of $Sc(OTf)_3$ was examined, and the results showed reducing the equivalent of $Sc(OTf)_3$ resulted in significant drop for the yield of **8a** (Table 1, entries 11-12). However, elevated reaction temperature (70 °C) could bring the yield back to 86% when 0.2 equivalent $Sc(OTf)_3$ was used (Table 1, entry 13).

Table 1. Optimization of reaction conditions.



12a **5a** **8a**

Entries[a]	LA (equiv.)	Temp/°C	Time/h	Y% [b]
1	TMSCl (1)	-78	12	39
2	$TiCl_4$ (2)	-78	12	NR
3	--	-78~r.t.	12	NR
4	$BF_3{\cdot}OEt_2$ (2)	-78	12	52
5	TMSOTf (2)	-78	12	44
6	$ZnCl_2$ (1)	-78	12	35
7	$Cu(OTf)_2$ (1)	-78~r.t.	12	65
8	$Zn(OTf)_2$ (1)	-78~r.t.	12	NR
9	$In(OTf)_2$ (1)	-78~r.t.	12	NR
10	$Sc(OTf)_3$ (1)	-78~r.t.	12	88
11	$Sc(OTf)_3$ (0.5)	-78~r.t.	12	62
12	$Sc(OTf)_3$ (0.2)	-78~r.t.	12	19
13	$Sc(OTf)_3$ (0.2)	r.t.~70	1	86

[a] Compound **12a** (0.5 mmol) was dissolved in dry THF (3 mL) under Ar atmosphere. DIBAL-H (1 M, 1.5 equiv) was dropped slowly. After being stirred at -78°C for 2 h, the mixture was quenched with EtOAc and potassium sodium tartrate, and extracted with EtOAc (20 mL×3). The combined organic layers were washed with brine, dried over $MgSO_4$, filtrated and concentrated, then obtained the unstable product **5a**, which was quickly uesed in next step without further purification. The above crude **5a**, **7a** (2.0 mmol) and LA were dissolved in dry THF (3 mL) at assigned reaction temperature. [b] Isolated yield (two steps). NR = No reaction.

With the optimized reaction conditions, we turned to investigate the scope and limitation of the nucleophilic addition of organozinc reagents **7** to *N,O*-acetal **5a**. Different substituted (methyl, fluoro, trifluoromethyl and chloro) benzyl zinc bromides were surveyed, as summarized in **Scheme 2**. In general, all these *ortho*-, *meta*- and *para*- substituted benzylzinc reagents could smoothly react with **5a**, affording the desired 1-benzylisoindolines **8b-8l** in moderate to excellent

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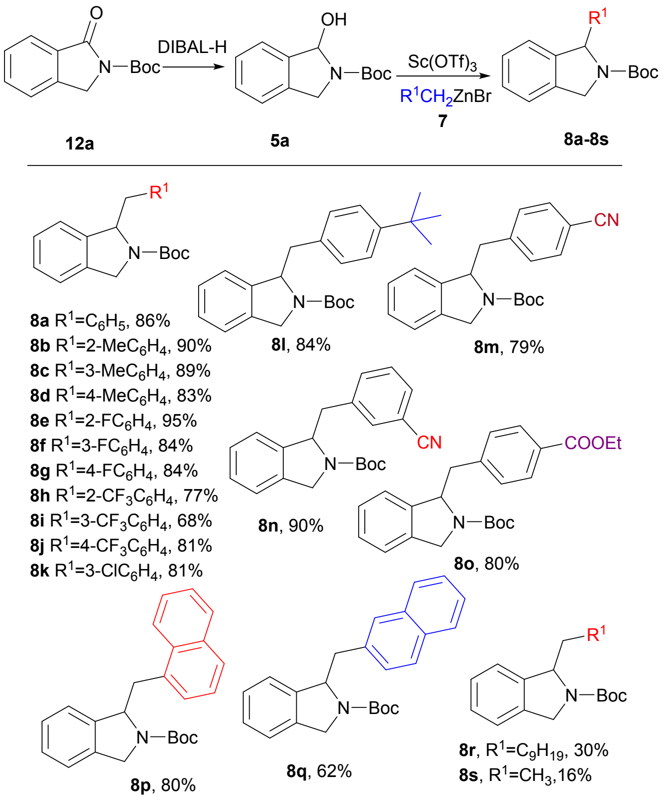
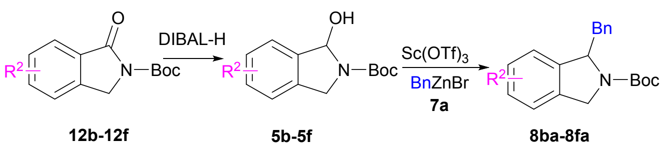
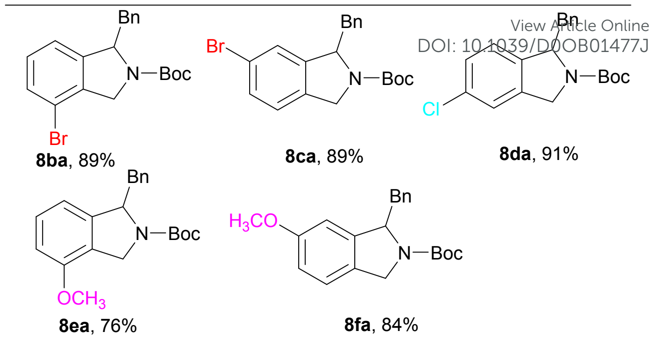
yields (68-95%). It is worth mentioning that cyano and carboxyl substituted benzylzinc bromides could successfully provide the products **8m, 8n** and **8o** in 79%, 90% and 80% yields, respectively. When 1-naphthalene zinc bromide and 2-naphthalene zinc bromide were used, the desired products **8p** and **8q** were produced in 80% and 62% yields, respectively. Notably, aliphatic zinc reagents, like *n*-decylzinc bromide and ethylzinc bromide could also afford the desired products **8r** and **8s**, albeit in low yields. Unfortunately, when ArZnBr and reformatsky type reagents[24] were examined, the reaction is very complicated and the desired product was not observed.



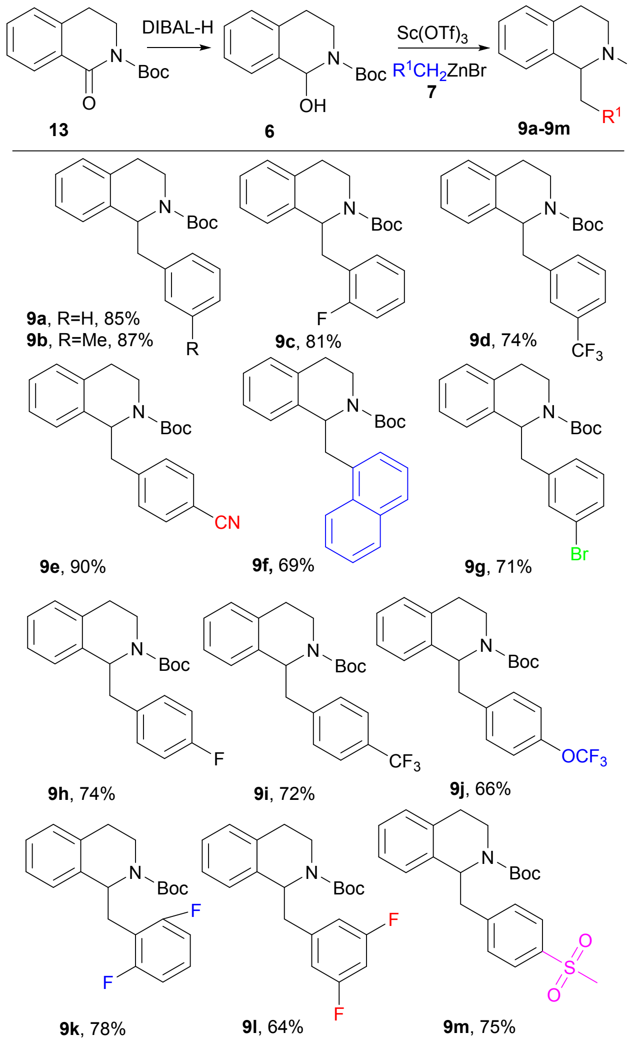
Scheme 2. Syntheses of 1-substituted isoindolines. [a] Compound **12a** (0.5 mmol) was dissolved in dry THF (3 mL) under Ar atmosphere. DIBAL-H (1 M, 1.5 equiv) was dropped slowly. After being stirred at -78°C for 2 h, the mixture was quenched with EtOAc and potassium sodium tartrate, and extracted with EtOAc (20 mL×3). The combined organic layers were washed with brine, dried over $MgSO_4$, filtrated and concentrated, then obtained the unstable product **5a**, which was quickly uesed in next step without further purification. The above crude **5a**, **7** (2.0 mmol) and $Sc(OTf)_3$ (0.1 mmol) were dissolved in dry THF (3 mL) at r.t.~70 °C for 1 h. [b] Isolated yield (two steps).

Then, the scope and limitation of *N,O*-acetals was also explored for this nucleophilic addition with benzylzinc bromide and the results are summarized in **Scheme 3**. The *N,O*-acetals **5b-5f** (obtained from reduction of **12b-12f**) with different substituents in benzene ring, including halogen and methoxy, could smoothly react with benzylzinc bromide to give the desired products **8ba**-**8fa** in excellent yields.





Scheme 3. The reactions of *N,O*-acetals **5b-5f** with benzylzinc bromide. [a] Compound **12b-12f** (0.5 mmol) was dissolved in dry THF (3 mL) under Ar atmosphere. DIBAL-H (1 M, 1.5 equiv) was dropped slowly. After being stirred at -78°C for 2 h, the mixture was quenched with EtOAc and potassium sodium tartrate, and extracted with EtOAc (20 mL×3). The combined organic layers were washed with brine, dried over $MgSO_4$, filtrated and concentrated, then obtained the unstable product **5b-5f**, which was quickly uesed in next step without further purification. The above crude **5b-5f**, **7a** (2.0 mmol) and $Sc(OTf)_3$ (0.1 mmol) were dissolved in dry THF (3 mL) at r.t.~70 °C for 1 h. [b] Isolated yield (two steps).

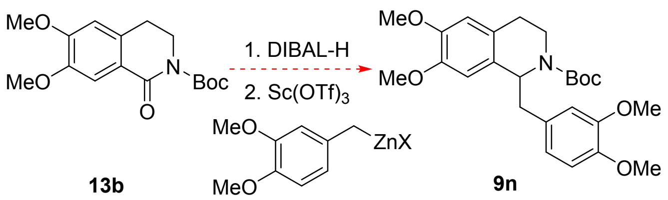


Scheme 4. Syntheses of 1-benzyl-tetrahydroisoquinolines. [a] Compound **13** (0.5 mmol) was dissolved in dry THF (3 mL) under Ar atmosphere. DIBAL-H (1 M, 1.5 equiv) was dropped slowly. After being stirred at -78°C for 2 h, the mixture was quenched with EtOAc and potassium sodium tartrate, and extracted with EtOAc (20 mL×3). The combined organic layers were washed with brine, dried over $MgSO_4$, filtrated and concentrated, then obtained the unstable product **6**, which was quickly uesed in next step without further purification. The above crude **6**, **7** (2.0 mmol) and $Sc(OTf)_3$ (0.1

mmol) were dissolved in dry THF (3 mL) at r.t.~70 °C for 1 h. [b] Isolated yield (two steps).

Next, we extended this method to construct 1-benzyl-tetrahydroisoquinolines. With $Sc(OTf)_3$ as the catalyst, various benzyl zinc bromides were examined and the results are summarized in **Scheme 4**. In general, most substituted benzyl zinc bromides could react smoothly to afford 1-benzyl-tetrahydroisoquinolines in moderate to excellent yields (**9a-9j**). Difluorobenzylzinc could also react well, and the yield of 2,6-difluoro-substituted benzyl zinc was slightly higher than that of 3,5-difluoro-substituted (78% for **9k**, 64% for **9l**), probably due to different electron cloud distribution. In addition, the sulfoxide substituted benzyl zinc bromide could also afford the corresponding product **9m** in 75% yield. The chemical structures of compounds **9a-9m** were unambiguously assigned based on the X-ray crystallographical analysis of compound **9i**.[25]

Finally, we turn our attention to synthesize the laudanosine (**1**) through this process (**Scheme 5**). However, when the **13b** was treated with DIBAL-H to obtain *N*,*O*-acetal **6b**, which was quickly uesed to react with freshly prepared (3,4-dimethoxybenzyl)zinc(II) bromide, the mixture became very complicated and no product **9n** was produced. The *N*,*O*-acetal **6** was also examined and the result is still fruitless. We suspect that the (3,4-dimethoxybenzyl)zinc(II) bromide reagent was not well formed. Although the (3,4-dimethoxybenzyl)zinc(II) bromide reagent has been emerged in one literature[28], the result is still negative. Combined with our previous results of organic zinc reagents[23b-d], we think that the methoxy substituted benzyl zinc reagents are very special and difficult for preparation.



Scheme 5. Syntheses of laudanosine (**1**).

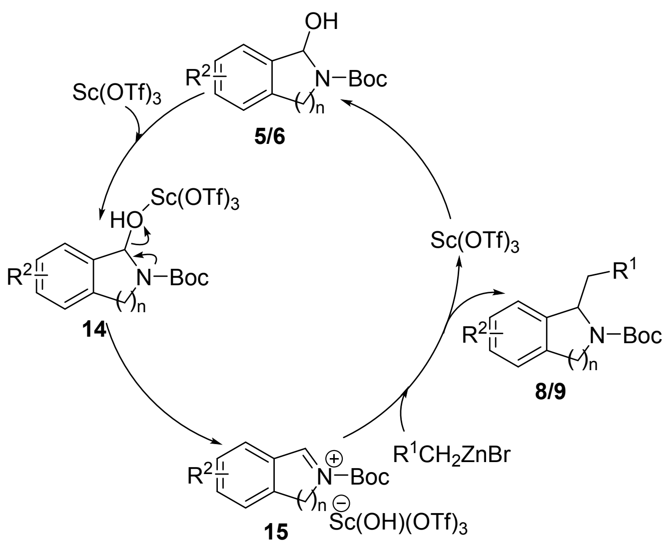


Figure 3. Proposed mechanism of the nucleophilic addition.

A possible mechanism for this nucleophilic addition was illustrated in **Figure 3**. First, the hydroxyl group in *N*,*O*-acetal **5/6** was removed by $Sc(OTf)_3$ catalyst to give iminium intermediate **15**.[26] Then, the nucleophilic benzylzinc attacked the C=N bond in **15** to afford the addition product **8/9** and release $Sc(OTf)_3$, which went back to participate in the reaction cycle.

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Conclusions

In summary, we have established a new and practical approach for the synthesis of 1-benzylisoindolines **8a-8s**, **8ba-8bf** and 1-benzyl-tetrahydroisoquinolines **9a-9m** through nucleophilic addition of *N*,O-acetals **5a-5f**, **6** with organozinc reagents **7**. $Sc(OTf)_3$ was found to be an effective catalyst for this tranformation, and a variety of substituted benzyl groups were successfully introduced to isoindoline and tetrahydroisoquinoline skeletons in moderate to excellent yields. To the best of our knowledge, the present process is not only the first direct method using organozinc reagents for the preparation of 1-benzylisoindolines and 1-benzyl-tetrahydroisoquinolines, but also the first example demonstrating that $Sc(OTf)_3$ could catalyze the addition or substitution reaction of *N*,*O*-acetals with organozinc reagents.

Experimental

General

General: THF was distilled from sodium/benzophenone, DCM was distilled from phosphorus pentoxide. Zinc dust was activated by being stirred for 15 min with HCl (2 N), followed by washing successively with water, ethyl alcohol, acetone, and ether. Reactions were monitored by TLC-FID. Flash chromatography was performed on silica gel (300-400 mesh) with petroleum/EtOAc as the eluent. HRMS was conducted on Thermo Scientific LTQ Orbitrap XL apparatus. IR spectra were measured using a film on a Fourier transform infrared spectrometer. NMR spectra were recorded at 400 MHz, and chemical shifts are reported in δ (ppm) referenced to an internal TMS standard for ^{1}H NMR and $CDCl_3$ (77.16 ppm) for $^{13}C\{^{1}H\}$ NMR. The heat source was oil bath.The Benzyl zinc bromide reagents were synthesized according to the known method.[23b]

General Procedure for the Synthesis of 8 or 9. Compound **12** or **13** (0.5 mmol) was dissolved in dry THF (3 mL) under Ar atmosphere. DIBAL-H (1 M, 1.5 equiv) was dropped slowly. After being stirred at -78°C for 2 h, the mixture was quenched with EtOAc and potassium sodium tartrate, and extracted with EtOAc (20 mL×3). The combined organic layers were washed with brine, dried over $MgSO_4$, filtrated and concentrated, then obtained the unstable product **5** or **6**, which was quickly uesed in next step without further purification. The above crude **5** or **6** and $Sc(OTf)_3$ (0.1 mmol, 49 mg) were dissolved in dry THF (3 mL) under Ar atmosphere. The freshly prepared substituted benzylzinc bromide reagent (2.0 mL, 1 M in THF) was added dropwise at room temperature, then stirred at 70°C for 1h.

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The mixture was quenched with a saturated aqueous solution of $NaHCO_3$ and extracted with EtOAc (20 mL × 3). The combined organic layers were washed with brine, dried over $MgSO_4$, filtrated and concentrated. The residue was purified by flash chromatography on silica gel (PE/EA=20:1~5:1) to give the desired product **8** or **9**.

***tert*-Butyl 1-benzylisoindoline-2-carboxylate 8a.** Yellow oil (133 mg, 86%); IR (film): ν_{max} 2974, 2925, 1698, 1454, 1397, 1366, 1172, 1110, 759, 700 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.21-7.15 (m, 3H), 7.14-7.09 (m, 2H), 7.07-6.97 (m, 1H), 6.96-6.81 (m, 3H), 5.36-5.31 (m, 0.45H), 5.28-5.20 (m, 0.55H), 4.63 (d, J = 14.8 Hz, 0.55H), 4.49 (d, J = 14.4 Hz, 0.45H), 4.12 (d, J = 14.8 Hz, 0.55H), 3.93 (d, J = 14.8 Hz, 0.45H), 3.36-3.28 (m, 0.45H), 3.26-3.19 (m, 1H), 3.12-3.04 (m, 0.55H), 1.59 (s, 4.95H), 1.54 (s, 4.05H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.3, 140.5, 140.2, 137.4, 137.2, 137.1, 137.0, 130.1, 130.0, 128.1, 127.8, 127.5, 127.0, 126.9, 126.4, 126.2, 123.2, 123.0, 122.6, 122.3, 80.0, 79.6, 64.1, 63.9, 52.2, 51.7, 41.5, 39.9, 28.8, 28.7 ppm; HRMS (ESI-Orbitrap) m/z: $[M + H]^+$ Calcd for $C_{20}H_{24}NO_2^+$: 310.1802, found: 310.1803.

***tert*-Butyl 1-(2-methylbenzyl)isoindoline-2-carboxylate 8b.** Yellow oil (146 mg, 90%); IR (film): ν_{max} 2974, 1698, 1463, 1395, 1366, 1171, 1106, 890, 750, 721 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.27-7.16 (m, 2H), 7.14-7.04 (m, 4H), 6.96-6.91 (m, 1H), 6.57-6.53 (m, 0.4H), 6.50-6.44 (m, 0.6H), 5.35-5.29 (m, 0.4H), 5.26-5.20 (m, 0.6H), 4.78 (d, J = 14.8 Hz, 0.6H), 4.63 (d, J = 14.8 Hz, 0.4H), 4.47 (d, J = 14.8 Hz, 0.6H), 4.34 (d, J = 14.8 Hz, 0.4H), 3.53-3.48 (m, 0.4H), 3.38-3.30 (m, 0.6H), 2.95-2.87 (m, 0.4H), 2.86-2.79 (m, 0.6H), 2.21-2.18 (m, 3H), 1.54 (s, 3.61H), 1.51 (s, 5.42H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.4, 154.3, 140.6, 140.3, 137.3, 137.2, 136.9, 135.9, 135.8, 131.3, 131.1, 130.3, 130.2, 127.5, 127.4, 126.7, 126.5, 125.8, 125.4, 123.7, 123.4, 122.6, 122.3, 80.1, 79.6, 63.2, 51.8, 51.4, 39.4, 37.9, 28.7, 19.8,19.7 ppm; HRMS (ESI-Orbitrap) m/z: $[M + H]^+$ Calcd for $C_{21}H_{26}NO_2^+$: 324.1958, found: 324.1958.

***tert*-Butyl 1-(3-methylbenzyl)isoindoline-2-carboxylate 8c.** Colourless oil (144 mg, 89%); IR (film): ν_{max} 2975, 2923, 1693, 1478, 1394, 1254, 1172, 1110, 772, 750 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.23-7.14 (m, 2H), 7.13-7.12 (m, 0.52H), 7.09-7.03 (m, 1H), 7.02-6.93 (m, 2H), 6.86-6.81 (m, 0.48H), 6.78-6.75 (m, 1H), 6.75-6.71 (m, 0.52H), 6.67-6.62 (m, 0.48H), 5.35-5.27 (m, 0.48H), 5.25-5.20 (m, 0.52H), 4.65 (d, J = 14.8 Hz, 0.52H), 4.49 (d, J = 14.8 Hz, 0.48H), 4.16 (d, J = 14.8 Hz, 0.52H), 3.97 (d, J = 14.8 Hz, 0.48H), 3.24-3.20 (m, 1H), 3.20-3.18 (m, 0.48H), 3.02-2.95 (m, 0.52H), 2.24-2.19 (m, 3H), 1.58 (s, 4.68H), 1.55 (s, 4.32H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.2, 154.2, 140.6, 137.5, 137.4, 137.2, 137.1, 137.1, 136.9, 130.9, 130.8, 128.0, 127.6, 127.5, 127.4, 127.1, 126.9, 123.2, 123.1, 122.5, 122.2, 79.9, 79.5, 64.1, 63.9, 52.1, 51.7, 41.5, 39.9, 28.7, 21.4, 21.3 ppm; HRMS (ESI-Orbitrap) m/z: $[M + H]^+$ Calcd for $C_{21}H_{26}NO_2^+$: 324.1958, found: 324.1957.

***tert*-Butyl 1-(4-methylbenzyl)isoindoline-2-carboxylate 8d.** Colourless oil (134 mg, 83%); IR (film): ν_{max} 2975, 2923, 1698, 1515, 1394, 1172, 1108, 900, 811, 750 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.23-7.16 (m, 2H), 7.15-7.10 (m, 0.55H), 7.06-7.02 (m, 0.45H), 6.99-6.95 (m, 1.45H), 6.95-6.90 (m, 1H), 6.88-6.84 (m, 0.55H), 6.82-6.76 (m, 2H), 5.32-5.17 (m, 1H), 4.63 (d, J = 14.8 Hz, 0.55H), 4.49 (d, J = 14.4 Hz, 0.45H), 4.12 (d, J = 14.8 Hz, 0.55H), 3.97 (d, J = 14.4 Hz, 0.45H), 3.29-3.15 (m, 1.45H), 3.06-2.99 (m, 0.55H), 2.29-2.24 (m, 3H), 1.59 (s, 4.95H), 1.54 (s, 4.05H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.3, 154.2, 140.6, 140.3, 137.4, 137.0, 135.8, 135.6, 134.0, 134.0, 129.9, 129.8, 128.8, 128.5, 127.4, 127.4, 126.9, 126.9, 123.2, 123.0, 122.5, 122.2, 79.9, 79.5, 64.2, 63.9, 52.2, 51.7, 41.0, 39.5, 28.7, 28.7, 21.1 ppm; HRMS (ESI-Orbitrap) m/z: $[M + H]^+$ Calcd for $C_{21}H_{26}NO_2^+$: 324.1958, found: 324.1958.

***tert*-Butyl 1-(2-fluorobenzyl)isoindoline-2-carboxylate 8e.** Colourless oil (156 mg, 95%); IR (film): ν_{max} 2975, 1698, 1492, 1395, 1366, 1231, 1171, 1111, 955, 754 cm^{-1}; ^1H NMR (400 MHz,$CDCl_3$,mixture of rotamers) δ 7.24-7.09 (m,4H), 6.97-6.84 (m, 4H), 5.41-5.36 (m, 0.38H), 5.33-5.27 (m, 0.62H), 4.69 (d, J = 14.8 Hz, 0.62H), 4.55 (d, J = 14.8 Hz, 0.38H), 4.25 (d, J = 14.8 Hz, 0.62H), 4.13 (d, J = 14.8 Hz, 0.38H), 3.39-3.32 (m, 0.38H), 3.29-3.20 (m, 1H), 3.16-3.10 (m, 0.62H), 1.54 (s, 9H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 161.6 (d, J = 244.3 Hz), 161.5 (d, J = 244.3 Hz), 162.7, 160.4, 160.3, 154.4, 154.3, 140.3, 139.8, 137.2, 137.0, 132.3, 132.3, 128.3 (d, J = 7.9 Hz), 128.1 (d, J = 7.9 Hz), 127.6, 127.5, 127.0, 126.9, 124.4 (d, J = 16.0 Hz), 124.3 (d, J = 16.0 Hz), 123.5, 123.5, 123.3, 123.1, 122.5, 122.2, 115.2 (d, J = 22.1 Hz), 115.0 (d, J = 22.2 Hz), 80.0, 79.7, 63.2, 63.1, 52.2, 51.6, 34.8, 33.0, 28.7, 28.6 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -117.0, -117.1 ppm; HRMS (ESI-Orbitrap) m/z: $[M + H]^+$ Calcd for $C_{20}H_{23}FNO_2^+$: 328.1707, found: 328.1705

***tert*-Butyl 1-(3-fluorobenzyl)isoindoline-2-carboxylate 8f.** Light-Yellow oil (138 mg, 84%); IR (film): ν_{max} 2975, 1698, 1588, 1488, 1396, 1254, 1172, 1111, 774, 750 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.25-7.17 (m,2H), 7.17-7.10 (m, 1H), 7.07-7.02 (m, 1.52H), 6.91-6.90 (m, 0.52H), 6.87-6.77 (m, 1.52H), 6.74-6.69 (m, 0.48H), 6.64-6.57 (m, 1.52H), 5.37-5.30 (m, 0.52H), 5.27-5.21 (m, 0.48H), 4.65 (d, J = 14.8 Hz, 0.48H), 4.49 (d, J = 14.4 Hz, 0.52H), 4.14 (d, J = 15.2 Hz, 0.48H), 3.95 (d, J = 14.4 Hz, 0.52H), 3.42-3.36 (m, 0.48H), 3.23-3.15 (m, 1H), 3.12-3.06 (m, 0.52H), 1.58 (s, 4.41H), 1.54 (s, 4.59H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 162.6 (d, J = 243.9 Hz), 162.5 (d, J = 243.2 Hz), 154.2, 154.2, 140.1, 139.8, 139.7, 137.3, 136.9, 129.5 (d, J = 8.1 Hz), 129.1 (d, J = 8.1 Hz), 127.7, 127.2, 127.1, 125.7, 125.6, 123.0, 122.9, 122.7, 122.4, 116.9 (d, J = 20.7 Hz), 116.7 (d, J = 20.6 Hz), 113.3 (d, J = 21.5 Hz), 113.1 (d, J = 21.3 Hz), 80.1, 79.7, 63.9, 63.6, 52.2, 51.8, 41.2, 39.6, 28.7, 28.6 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -114.1 (dd, J = 8.3, 15.4 Hz), -114.7 (dd, J = 8.3, 15.4 Hz) ppm; HRMS (ESI-Orbitrap) m/z: $[M + H]^+$ Calcd for $C_{20}H_{23}FNO_2^+$: 348.1707, found: 328.1704.

***tert*-Butyl 1-(4-fluorobenzyl)isoindoline-2-carboxylate 8g.** Colourless oil (138 mg, 84%); IR (film): ν_{max} 2975, 1698, 1509, 1395, 1366, 1221, 1173, 1110, 825, 750 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.25-7.18(m,2H), 7.15-7.07 (m, 0.51H), 7.07-7.02 (m, 1H), 6.96-6.90 (m, 0.49H), 6.85-6.77 (m, 4H), 5.35-5.29 (m, 0.49H), 5.25-5.20 (m, 0.51H), 4.62 (d, J = 14.8 Hz, 0.51H), 4.48 (d, J = 14.8 Hz, 0.5H), 4.05 (d, J = 14.8 Hz, 0.49H), 3.90 (d, J = 14.8 Hz, 0.52H), 3.41-3.34 (m, 0.49H), 3.15-3.11 (m, 1.51H), 1.59 (s, 4.51H), 1.54 (s, 4.59H) ppm; ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 161.7 (d, J = 242.9 Hz), 154.2, 137.4, 131.4,131.3,131.3, 127.6, 127.1, 127.0, 122.9 (d, J = 14.9 Hz), 122.5 (d, J = 27.9 Hz), 114.9 (d, J = 20.7 Hz), 114.6 (d, J = 20.8 Hz), 80.1, 79.7, 64.0, 63.8, 52.3, 51.8, 40.4, 39.0, 28.8, 28.7 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -117.0, -117.4 ppm; HRMS (ESI-Orbitrap) m/z: [M + H]$^+$ Calcd for $C_{20}H_{23}FNO_2^+$: 328.1707, found: 328.1707.

***tert*-Butyl 1-(2-(trifluoromethyl)benzyl)isoindoline-2-carboxylate 8h.** Colourless oil (145 mg, 77%); IR (film): ν_{max} 2978, 1698, 1392, 1314, 1164, 1111, 1060, 1039, 767, 754 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.68-7.67 (m, 0.7H), 7.63-7.59 (m, 0.3H), 7.47-7.42 (m, 1.3H), 7.39-7.35 (m, 0.7H), 7.31-7.26 (m, 1H), 7.26-7.21 (m, 1H), 7.20-7.17 (m, 1H), 7.15-7.10 (m, 0.7H), 7.07-7.03 (m, 0.3H), 6.73-6.68 (m, 0.7H), 6.49-6.45 (m, 0.3H), 5.47-5.40 (m, 0.3H), 5.37-5.32 (m, 0.7H), 4.90 (d, J = 15.2 Hz, 0.7H), 4.71 (d, J = 15.2 Hz, 0.3H), 4.63-4.57 (m, 1H), 3.68-3.50 (m, 0.3H), 3.24-3.15 (m, 0.7H), 3.14-3.05 (m, 1H), 1.49 (s, 2.7H), 1.29 (s, 7.3H) ppm; ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ154.7, 154.4, 141.0, 140.2, 137.0, 133.2, 132.3, 132.3, 131.7, 131.5, 129.4 (q, J = 29.1 Hz), 129.3 (q, J = 29.6 Hz), 127.7 (d, J = 8.4Hz), 126.9 (d, J = 8.6 Hz), 126.7, 126.5, 126.2, 126.1, 123.3, 123.2, 123.1, 122.7, 122.4, 80.0, 79.8, 63.9, 63.8, 51.8, 51.3, 40.1, 38.0, 28.6, 28.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -58.5, -58.9 ppm; HRMS (ESI-Orbitrap) m/z: [M + H]$^+$ Calcd for $C_{20}H_{23}F_3NO_2^+$: 378.1675, found: 378.1677.

***tert*-Butyl 1-(3-(trifluoromethyl)benzyl)isoindoline-2-carboxylate 8i.** Colourless oil (128 mg, 68%); IR (film): ν_{max} 2977, 1698, 1396, 1367, 1326, 1167, 1124, 1075, 750, 705 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.44-7.36 (m, 1H), 7.32-7.25 (m, 1H), 7.23-7.15 (m, 2H), 7.13-7.08 (m, 2H), 7.03-6.99 (m, 0.58H), 6.97-6.92 (m, 1H), 5.41-5.35 (m, 0.58H), 5.30-5.25 (m, 0.42H), 4.62 (d, J = 14.8 Hz, 0.42H), 4.45 (d, J = 14.8 Hz, 0.58H), 4.01 (d, J = 14.4 Hz, 0.42H), 3.79 (d, J = 14.8 Hz, 0.58H), 3.58-3.52 (m, 0.58H), 3.23-3.13 (m, 1.42H), 1.59 (s, 3.78H), 1.54 (s, 5.22H) ppm; ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ154.3, 154.2, 139.5, 138.1, 137.4, 137.0, 133.4, 133.3, 129.4 (q, J = 29.5 Hz), 129.3 (q, J = 29.6 Hz), 128.6, 128.1, 127.9, 127.8, 127.3, 127.2, 126.9 (d, J = 3.2 Hz), 126.7 (d, J = 3.4 Hz), 125.6, 125.5, 123.3, 123.1, 123.1, 122.9, 122.8, 122.7, 122.4, 80.3, 79.9, 63.9, 63.6, 52.3, 51.8, 41.3, 39.5, 28.7, 28.6 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -57.9, -58.0 ppm; HRMS (ESI-Orbitrap) m/z: [M + H]$^+$ Calcd for $C_{21}H_{23}F_3NO_2^+$: 378.1675, found: 378.1672.

***tert*-Butyl 1-(4-(trifluoromethyl)benzyl)isoindoline-2-carboxylate 8j.** Colourless oil (153 mg, 81%); IR (film): ν_{max} 2977, 1698, 1617, 1394, 1325, 1288, 1166, 1108, 836, 751 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.43-7.34 (m,2H), 7.27-7.22 (m, 2H), 7.15-6.94 (m, 4H), 5.40-5.35 (m, 0.53H), 5.31-5.26 (m, 0.47H), 4.64 (d, J = 14.8 Hz, 0.47H), 4.49 (d, J = 14.8 Hz, 0.53H), 4.05 (d, J = 14.8 Hz, 0.47H), 3.91 (d, J = 14.8 Hz, 0.53H), 3.50-3.42 (m, 0.53H), 3.25-3.19 (m, 1.47H), 1.58 (s, 4.22H), 1.54 (s, 4.78H) ppm; ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, mixture of rotamers) $\delta\delta$ 154.3, 154.2, 141.5, 141.4, 139.9, 139.7, 137.3, 137.0, 130.4, 130.2, 128.6 (q, J = 32.3 Hz), 128.6 (q, J = 32.4 Hz), 127.8, 127.3, 127.2, 125.8, 125.7, 125.0 (d, J = 3.2 Hz), 124.7 (d, J = 3.3 Hz), 123.1, 123.0, 122.8, 122.5, 80.3, 79.9, 77.5, 77.2, 76.8, 63.9, 63.7, 52.3, 51.8, 41.3, 39.7, 28.8, 28.7 ppm; ^{19}F NMR (376 MHz, CDCl3, mixture of rotamers) δ -57.5, -57.6 ppm; HRMS (ESI-Orbitrap) m/z: [M + H]$^+$ Calcd for $C_{21}H_{23}F_3NO_2^+$: 378.1675, found: 378.1674.

***tert*-Butyl 1-(3-chlorobenzyl)isoindoline-2-carboxylate 8k.** Colourless oil (139 mg, 81%); IR (film): ν_{max} 2975, 2928, 1698, 1598, 1477, 1395, 1172, 1078, 961, 807 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.29-7.18 (m, 2.46H), 7.17-7.09 (m, 2H), 7.07-7.04 (m, 1H), 7.03-6.99 (m, 0.54H), 6.93-6.89 (m, 1H), 6.84-6.80 (m, 0.46H), 6.70-6.66 (m, 0.54H), 5.36-5.29 (m, 0.54H), 5.26-5.21 (m, 0.46H), 4.66 (d, J = 14.8 Hz, 0.46H), 4.49 (d, J = 14.4 Hz, 0.54H), 4.14 (d, J = 14.4 Hz, 0.46H), 3.92 (d, J = 14.8 Hz, 0.54H), 3.42-3.35 (m, 0.54H), 3.18-3.11 (m, 1H), 3.10-3.04 (m, 0.46H), 1.58 (s, 4.14H), 1.55 (s, 4.86H) ppm; ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.2, 140.1, 139.8, 139.3, 139.3, 137.4, 137.0, 133.9, 133.6, 130.2, 130.0, 129.4, 129.0, 128.2, 128.1, 127.8, 127.3, 127.2, 126.7, 126.5, 123.0, 122.9, 122.7, 122.4, 80.2, 79.8, 63.9, 63.6, 52.3, 51.8, 41.3, 39.5, 28.8, 28.7 ppm; HRMS (ESI-Orbitrap) m/z: [M + H]$^+$ Calcd for $C_{20}H_{23}ClNO_2^+$: 344.1412, found: 344.1415.

***tert*-Butyl 1-(4-(tert-butyl)benzyl)isoindoline-2-carboxylate 8l.** Colourless oil (154 mg, 84%); IR (film): ν_{max} 2964, 2866, 1699, 1464, 1395, 1365, 1173, 1108, 830, 772 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.24-7.14 (m, 4.6H), 7.09-7.04 (m, 0.4H), 6.92-6.83 (m, 3H), 5.32-5.28 (m, 0.4H), 5.24-5.19 (m, 0.6H), 4.66 (d, J = 14.8 Hz, 0.6H), 4.52 (d, J = 14.8 Hz, 0.4H), 4.16 (d, J = 14.8 Hz, 0.6H), 4.02 (d, J = 14.8 Hz, 0.4H), 3.27-3.14 (m, 1.4H), 3.04-2.97 (m, 0.6H), 1.56 (s, 5.4H), 1.54 (s, 3.6H), 1.29-1.26 (m, 9H) ppm; ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.3, 154.2, 149.2, 149.0, 140.8, 140.5, 137.4, 137.0, 134.2, 129.7, 129.6, 127.5, 127.4, 126.9, 125.1, 124.7, 123.3, 123.0, 122.5, 122.2, 79.9, 79.5, 64.2, 64.0, 52.1, 51.7, 41.2, 39.7, 34.4, 31.5, 28.7 ppm; HRMS (ESI-Orbitrap) m/z: [M + H]$^+$ Calcd for $C_{24}H_{32}NO_2^+$: 366.2428, found: 366.2430.

***tert*-Butyl 1-(4-cyanobenzyl)isoindoline-2-carboxylate 8m.** Colourless oil (132 mg, 79%); IR (film): ν_{max} 2975, 2929, 2227, 1697, 1608, 1394, 1171, 1109, 823, 750 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.44-7.36 (m, 2H), 7.27-7.21 (m, 2H), 7.15-7.01 (m, 2H), 6.98-6.93 (m, 2H), 5.40-5.27 (m, 1H), 4.61 (d, J = 15.2 Hz, 0.42H), 4.48 (d, J = 14.4 Hz, 0.58H), 3.98 (d, J = 14.8 Hz, 0.42H), 3.86 (d, J = 14.4 Hz, 0.58H), 3.58-3.47 (m, 0.58H), 3.34-3.25 (m, 0.42H), 3.21-3.13 (m, 1H), 1.59 (s, 3.76H), 1.54 (s, 5.22H) ppm; ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.1, 153.9, 143.0, 137.1, 131.6, 131.4, 130.6, 130.6, 127.8, 127.3, 127.2, 122.7, 122.6, 122.6, 122.4, 110.2, 110.1, 80.2, 79.8, 63.6, 63.4, 52.2, 51.8, 41.3, 39.8, 28.6, 28.5 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{21}H_{23}N_2O_2^+$: 335.1754, found: 335.1756.

***tert*-Butyl 1-(2-cyanobenzyl)isoindoline-2-carboxylate 8n.** White solid (150 mg, 90%); m.p. 87-89ºC; IR (film): v_{max} 2975, 2224, 1697, 1449, 1392, 1367, 1169, 1110, 957, 762 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.63-7.51 (m, 1H), 7.47-7.42 (m, 1H), 7.35-7.27 (m, 2H), 7.24-7.20 (m,2H), 7.18-7.14 (m, 1H), 7.08-7.02 (m, 1H), 5.45-5.39 (m, 1H), 4.76 (d, *J* = 15.2 Hz, 0.55H), 4.62 (d, *J* = 14.8 Hz, 0.45H), 4.33-4.25 (m, 1H), 3.50-3.42 (m, 1H), 3.42-3.38 (m, 0.45H), 3.32-3.27 (m, 0.55H), 1.52 (s, 4.05H), 1.46 (s, 4.95H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.2, 141.5, 141.4, 139.5, 136.8, 133.0, 132.8, 132.6, 132.4, 132.3, 132.2, 130.8, 129.8, 127.8, 127.2, 127.0, 127.0, 126.8, 123.2, 123.0, 122.5, 122.3, 118.0, 113.6, 80.2, 79.8, 63.5, 63.4, 52.0, 51.6, 40.2, 39.0, 35.6, 28.5, 28.4 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{21}H_{23}N_2O_2^+$: 335.1754, found: 335.1755.

***t*ert-Butyl 1-(4-(ethoxycarbonyl)benzyl)isoindoline-2-carboxylate 8o.** Colourless oil (153 mg, 80%); IR (film): v_{max} 2977, 1718, 1698, 1394, 1275, 1176, 1104, 1022, 761, 709 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.87-7.82 (m, 1H), 7.81-7.77 (m, 1H), 7.25-7.16 (m, 2H), 7.15-7.08 (m, 0.46H), 7.07-7.02 (m, 1H), 7.00-6.95 (m, 1H), 6.95-6.92 (m, 1H), 6.92-6.87 (m, 0.54H), 5.40-5.34 (m, 0.54H), 5.31-5.25 (m, 0.46H), 4.62 (d, *J* = 14.8 Hz, 0.46H), 4.48 (d, *J* = 14.4 Hz, 0.54H), 4.37-4.30 (m, 2H), 4.05 (d, *J* = 14.4 Hz, 0.46H), 3.89 (d, *J* = 14.8 Hz, 0.54H), 3.50-3.44 (m, 0.54H), 3.26-3.21 (m, 1H), 3.21-3.18 (m, 0.46H), 1.60 (s, 4.14H), 1.55 (s, 4.86H), 1.41-1.34 (m, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 166.9, 166.8, 154.3, 154.2, 142.8, 142.6, 140.0, 139.7, 137.3, 137.0, 130.1, 130.0, 129.4, 129.1, 128.7, 128.5, 127.8, 127.2, 127.1, 123.0, 122.9, 122.7, 122.5, 80.2, 79.8, 63.9, 63.7, 61.0, 52.3, 51.8, 41.4, 39.9, 28.8, 28.7, 14.5 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{23}H_{28}NO_4^+$: 382.2013, found: 382.2013.

***tert*-Butyl 1-(naphthalen-1-ylmethyl)isoindoline-2-carboxylate 8p.** Whitefoam (144 mg, 80%); IR (film): v_{max} 2975, 1695, 1453, 1394, 1366, 1169, 1108, 791, 774, 753 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 8.52-8.48 (m, 0.40H), 8.38-8.31 (m, 0.60H), 7.92-7.82 (m, 1H), 7.80-1.72 (m, 1H), 7.57-7.53 (m, 0.40H), 7.52-7.50 (m, 1H), 7.49-7.45 (m, 0.60H), 7.39-7.26 (m, 1H), 7.24-7.10 (m, 2H), 7.02-6.94 (m, 2H), 6.36-6.29 (m, 0.40H), 6.20-6.15 (m, 0.60H), 5.55-5.48 (m, 0.40H), 5.47-5.41 (m, 0.60H), 4.82 (d, *J* = 14.8 Hz, 0.6H), 4.62 (d, *J* = 14.4 Hz, 0.4H), 4.55 (d, *J* = 14.4 Hz, 0.6H), 4.32 (d, *J* = 14.8 Hz, 0.4H), 4.13-4.07 (m, 0.4H), 3.97-3.91 (m, 0.6H), 3.23-2.93 (m, 1H), 1.56 (s, 9H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.4, 136.8, 133.9, 133.8, 133.7, 129.1, 128.9, 128.4, 127.5, 127.4, 127.3, 126.5, 126.4, 126.3, 125.8, 125.7, 125.6, 125.4, 125.0, 124.4, 124.2, 124.0, 122.5, 122.3, 80.5, 79.7, 63.1, 62.8, 51.9, 51.5, 40.2, 38.4, 28.8 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{24}H_{26}NO_2^+$: 360.1958, found: 360.1959.

***t*ert-Butyl 1-(naphthalen-2-ylmethyl)isoindoline-2-carboxylate 8q.** Whitefoam (111 mg, 62%); IR (film): v_{max} 2974, 1696, 1508, 1395, 1366, 1255, 1169, 1110, 816, 747 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.86-7.74 (m, 2H), 7.69-7.65 (m, 1H), 7.61-7.57 (m, 0.50H), 7.45-7.37 (m, 4H), 7.23-7.18 (m, 1H), 7.17-7.14 (m, 0.50H), 7.13-7.09 (m, 1H), 7.04-6.97 (m, 1.50H), 6.87-6.82 (m, 0.50H), 5.47-5.40 (m, 0.50H), 5.36-5.31 (m, 0.50H), 4.65 (d, *J* = 14.8 Hz, 0.50H), 4.48 (d, *J* = 14.8 Hz, 0.50H), 4.13 (d, *J* = 15.2 Hz, 0.50H), 3.91 (d, *J* = 15.2 Hz, 0.50H), 3.47-3.38 (m, 1H), 3.25-3.16 (m, 1H), 1.61 (s, 4.50H), 1.56 (s, 4.50H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.4, 135.0, 134.9, 133.4, 132.2, 128.7, 128.6, 128.2, 128.0, 127.7, 127.6, 127.5, 127.2, 127.1, 127.0, 126.6, 126.0, 125.9, 125.7,125.5, 125.3, 123.2, 123.1, 122.7, 122.4, 80.1, 79.7, 64.2, 64.0, 52.3, 51.8, 41.8, 40.1, 38.2, 28.8, 28.7 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{24}H_{26}NO_2^+$: 360.1958, found: 360.1955.

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***tert*-Butyl 1-decylisoindoline-2-carboxylate 8r.** Light-yellow oil (54 mg, 30%); IR (film): v_{max} 2925, 1701, 1464, 1392, 1366, 1288, 1255, 1173, 1107, 748 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.28-7.15 (m, 4H), 5.15-5.00 (m, 1H), 4.80-4.65 (m, 1H), 4.60-4.50 (m, 1H), 2.17-1.98 (m, 1H), 1.84-1.69 (m, 1H), 1.52 (s, 9H), 1.30-1.15 (m, 16H), 0.93-0.81 (m, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.7, 154.4, 141.5, 141.4, 137.2, 136.9, 127.3, 122.7, 122.6, 122.5, 122.4, 79.7, 79.5, 63.1, 52.7, 52.4, 35.4, 34.4, 32.0, 29.9, 29.7, 29.4, 28.7, 23.8, 23.6, 22.8, 14.2 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + Na]^+$ Calcd for $C_{23}H_{37}NO_2Na^+$: 382.2717, found: 382.2713.

***tert*-Butyl 1-ethylisoindoline-2-carboxylate 8s.** Colourless oil (20 mg, 16%); IR (film): v_{max} 2969, 2875, 1698, 1455, 1393, 1289, 1179, 1113, 876, 746 cm^{-1};^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.21-7.09 (m,4H), 5.10-5.04 (m, 0.42H), 5.02-4.95 (m, 0.58H), 4.71 (d, *J* = 14.8 Hz, 0.58H), 4.63 (d, *J* = 15.2 Hz, 0.42H), 4.52-4.45 (m, 1H), 2.18-2.01 (m, 1H), 1.84-1.62 (m, 1H), 1.45 (s, 9H),0.66-0.57 (m, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.7, 154.4, 141.0, 140.9, 137.3, 137.1, 127.4, 127.4, 122.7, 122.6, 122.5, 122.4, 79.7, 79.5, 63.8, 52.9, 52.6, 28.7, 28.1, 27.0, 7.8, 7.7, 1.2 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{15}H_{22}NO_2^+$: 248.1645, found: 248.1641.

***tert*-Butyl 1-benzyl-4-bromoisoindoline-2-carboxylate 8ba.** Colourless oil (173 mg, 89%); IR (film): v_{max} 2975, 1699, 1576, 1474, 1394, 1366, 1170, 1111, 776, 701 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.36-7.31 (m, 1H), 7.22-7.13 (m, 3H), 7.09-7.01 (m, 1H), 6.97-6.88 (m, 2H), 6.87-6.83 (m, 0.50H), 6.72-6.67 (m, 0.50H), 5.42-5.37 (m, 0.5H), 5.32-5.28 (m, 0.5H), 4.66 (d, *J* = 15.6 Hz, 0.5H), 4.47 (d, *J* = 15.2 Hz, 0.5H), 4.12 (d, *J* = 15.2 Hz, 0.5H), 3.92 (d, *J* = 15.2 Hz, 0.5H), 3.25-3.20 (m, 1.5H), 3.05-2.96 (m, 0.5H), 1.58 (s, 4.49H), 1.55(s, 4.51H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.0, 142.3, 142.1, 138.1, 137.8, 136.7, 130.5, 130.5, 130.0, 129.8, 128.9, 128.8, 128.3, 128.0, 126.7, 126.5, 122.0, 121.9, 117.4, 117.1, 80.2, 80.0, 65.2, 65.0, 53.4, 53.1, 41.6, 40.1, 28.7 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{20}H_{23}BrNO_2^+$: 388.0907, 390.0886, found: 388.0907, 390.0886.

***tert*-Butyl 1-benzyl-6-bromoisoindoline-2-carboxylate 8ca.** Colourless oil (173 mg, 89%); IR (film): v_{max} 2975, 1698, 1476, 1454, 1396, 1366, 1168, 1112, 812, 701 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.34-7.29 (m, 1H), 7.18-7.17 (m, 1H), 7.16-7.11 (m, 2H), 7.02-6.97 (m, 1H), 6.97-6.84 (m, 3H), 5.31-5.27 (m, 0.43H), 5.23-5.17 (m, 0.57H), 4.55 (d, *J* = 15.2 Hz, 0.57H), 4.41 (d, *J* = 14.8 Hz, 0.43H), 4.00 (d, *J* = 14.8 Hz, 0.57H), 3.83 (d, *J* = 14.8 Hz, 0.43H), 3.36-3.30 (m, 0.44H), 3.18-3.14 (m, 1H), 3.11-3.04 (m, 0.57H), 1.58 (s, 5.13H), 1.54 (s,3.87H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.0, 142.8, 142.5, 136.6, 136.5, 136.4, 136.1, 130.6, 130.0, 129.9, 128.2, 127.9, 126.6, 126.4, 126.3, 126.2, 124.1, 123.8, 120.7, 120.6, 80.2, 79.8, 63.8, 63.6, 51.8, 51.4, 41.3, 39.8, 28.7, 28.6 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{20}H_{23}BrNO_2^+$: 388.0907, 390.0886, found: 388.0901, 390.0880.

***tert*-Butyl 1-benzyl-5-chloroisoindoline-2-carboxylate 8da.** Colourless oil (156 mg, 91%); IR (film): v_{max}2976, 1697, 1478, 1394, 1258, 1169, 1076, 865, 738, 701 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.19-7.12 (m, 4H), 7.11-6.98 (m, 1H), 6.93-6.72 (m, 3H), 5.31-5.23 (m, 0.47H), 5.22-5.15 (m, 0.53H), 4.60 (d, *J* = 14.8 Hz, 0.53H), 4.45 (d, *J* = 14.8 Hz, 0.47H), 4.09 (d, *J* = 14.8 Hz, 0.53H), 3.91 (d, *J* = 14.8 Hz, 0.47H), 3.31-3.24 (m, 0.43H), 3.24-3.16 (m, 1H), 3.08-3.00 (m, 0.57H), 1.59 (s, 4.77H), 1.54 (s, 4.23H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.0, 139.3, 139.0, 138.9, 138.7, 136.8, 136.7, 133.4, 133.3, 130.0, 129.9, 128.2, 128.0, 127.3, 127.2, 126.6, 126.4, 124.3, 124.2, 122.8, 122.5, 80.2, 79.8, 63.7, 63.5, 51.8, 51.4, 41.3, 39.8, 28.7, 28.6 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{20}H_{23}ClNO_2^+$: 344.1412, found: 344.1412.

***tert*-Butyl 1-benzyl-4-methoxyisoindoline-2-carboxylate 8ea.** Colourless oil (129 mg, 76%); IR (film): v_{max} 2973, 1697, 1615, 1489, 1441, 1396, 1297, 1111, 775, 701 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.21-7.10 (m, 4H), 6.99-6.90 (m, 2H), 6.72-6.65 (m, 1H), 6.60-6.53 (m, 0.55H), 6.46-6.40 (m,0.45H), 5.35-5.27 (m, 0.55H), 5.25-5.19 (m,0.55H), 4.62 (d, *J* = 14.8 Hz, 0.45H), 4.47 (d, *J* = 14.8 Hz, 0.55H), 4.08 (d, *J* = 14.8 Hz, 0.45H), 3.92 (d, *J* = 14.8 Hz, 0.55H), 3.77 (s, 3H), 3.27-3.20 (m, 1.55H), 3.07-3.00 (m, 0.45H), 1.58 (s, 4.05H), 1.54 (s, 4.95H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.7, 154.4, 154.3, 142.3, 142.1, 137.3, 130.1, 130.0, 128.7, 128.6, 128.2, 127.9, 126.4, 126.3, 125.5, 125.2, 115.5, 115.2, 108.9, 108.7, 79.9, 79.6, 64.7, 64.4, 55.4, 55.3, 50.2, 49.8, 41.5, 40.0, 28.8, 28.7 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{21}H_{26}NO_3^+$: 340.1907, found: 340.1908.

***tert*-Butyl 1-benzyl-6-methoxyisoindoline-2-carboxylate 8fa.** Colourless oil (143 mg, 84%); IR (film): v_{max}2974, 1697, 1498, 1398, 1271, 1207, 1170, 1109, 881, 772 cm^{-1};^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.20-7.13 (m, 3H), 7.03-6.93 (m, 3H), 6.77-6.73 (m, 1H), 6.45-6.40 (m, 0.47H), 6.33-6.28 (m, 0.53H), 5.31-5.24 (m, 0.47H), 5.21-5.16 (m, 0.53H), 4.57 (d, *J* = 14.4 Hz, 0.53H), 4.43 (d, *J* = 14.0 Hz, 0.47H), 4.09 (d, *J* = 14.0 Hz, 0.53H), 3.92 (d, *J* = 14.0 Hz, 0.47H), 3.72-3.68 (m, 3H), 3.26-3.21 (m, 1.47H), 3.05-2.96 (m, 0.53H), 1.59 (s, 5.13H), 1.54 (s, 3.87H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 159.0, 158.9, 154.2, 141.9, 141.6, 137.3, 137.2, 130.1, 130.0, 128.2, 127.9, 126.5, 126.3, 123.2, 123.0, 114.2, 114.0, 108.1, 108.0, 79.9, 79.5, 64.3, 64.0, 55.4, 51.6, 51.2, 41.5, 40.0, 28.7, 28.7 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + H]^+$ Calcd for $C_{21}H_{26}NO_3^+$: 340.1907, found: 340.1905.

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***tert*-Butyl 1-benzyl-3,4-dihydroisoquinoline-2(1H)-carboxylate 9a.** Whitefoam (137 mg, 85%); IR (film): v_{max}2973, 2927, 1690, 1417, 1390, 1364, 1164, 1118, 753cm^{-1}; ^1H NMR (400 MHz, $DMSO^{-d6}$, 80 °C) δ 7.50-6.90 (m, 9H), 5.36-5.14 (m, 1H), 3.94 (brs, 1H), 3.36-3.26 (m, 1H), 3.07-3.02 (m, 2H), 2.84-2.66 (m, 2H), 1.26 (s, 9H) ppm; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.31-7.26 (m, 1H), 7.25-7.23 (m, 0.7H), 7.23-7.19 (m, 1H), 7.19-7.11 (m, 4H), 7.11-7.03 (m, 2H), 6.91-6.86 (m, 0.3H), 5.37 (dd, *J* = 7.2, 6.4 Hz, 0.3H), 5.22 (dd, *J* = 8.4, 5.6 Hz, 0.7H), 4.25-4.17 (m, 0.7H), 3.83-3.75 (m, 0.3H), 3.37-3.33 (m, 0.3H), 3.32-3.24 (m, 0.7H), 3.09-2.97 (m, 2H), 2.97-2.87 (m, 0.7H), 2.84-2.76 (m, 0.3H), 2.74-2.67 (m, 0.7H), 2.65-2.57 (m, 0.3H), 1.40 (s, 2.7H), 1.21 (s, 6.3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.7, 154.6, 138.7, 138.4, 137.2, 137.1, 134.9, 134.7, 129.8, 129.2, 128.5, 128.4, 128.2, 127.7, 127.3, 126.8, 126.7, 126.5, 126.3, 126.0, 79.7, 79.6, 56.9, 55.8, 43.1, 42.9, 39.4, 37.0, 28.8, 28.7, 28.5, 28.2 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + Na]^+$ Calcd for $C_{21}H_{25}NO_2Na^+$: 346.1778, found: 346.1776.

***tert*-Butyl 1-(3-methylbenzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9b.** Whitefoam (147 mg, 87%); IR (film): v_{max}2974, 2927, 1692, 1418, 1365, 1165, 1119, 770, 699 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.19-6.98 (m, 6H), 6.95-6.83 (m, 2H), 5.35 (dd, *J* = 7.2, 6.4 Hz, 0.28H), 5.21 (dd, *J* = 8.8, 5.2 Hz, 0.72H), 4.26-4.17 (m, 0.72H), 3.85-3.75 (m, 0.28H), 3.38-3.35 (m, 0.28H), 3.33-3.25 (m, 0.72H), 3.07-2.96 (m, 2H), 2.95-2.87 (m, 0.72H), 2.84-2.77 (m, 0.28H), 2.75-2.68 (m, 0.72H), 2.66-2.58 (m, 0.28H), 2.32 (s, 2H), 2.28 (s, 1H), 1.41 (s, 2.52H), 1.20 (s, 6.48H) ppm ; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.8, 154.6, 138.7, 138.3, 137.8, 137.6, 137.4, 137.2, 134.8, 134.7, 130.6, 130.5, 129.2, 128.5, 128.4, 128.0, 127.8, 127.4, 127.2, 127.1, 126.9, 126.8, 126.7, 126.6, 126.0, 125.9, 79.6, 57.0, 55.9, 43.1, 42.8, 39.5, 37.0, 28.8, 28.7, 28.6, 28.2, 21.5 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + Na]^+$ Calcd for $C_{22}H_{27}NO_2N_a^+$: 360.1934, found: 360.1936.

***tert*-Butyl 1-(2-fluorobenzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9c.** Whitefoam (138 mg, 81%); IR (film): v_{max} 2975, 2929, 1692, 1492, 1419, 1365, 1230, 1164, 1120, 754 cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.24-7.21 (m, 1H), 7.20-7.18 (m, 2H), 7.17-7.12 (m, 2H), 7.09-6.99 (m, 3H), 5.45 (dd, *J* = 8.8, 5.6 Hz, 0.24H), 5.34 (dd, *J* = 10.4, 4.0 Hz, 0.76H), 4.33-4.25 (m, 0.76H), 3.95-3.87 (m, 0.24H), 3.46-3.40 (m, 0.24H), 3.35-3.26 (m, 0.76H), 3.20-3.13 (m, 1H), 3.09-3.03 (m, 0.24H), 3.02-2.97 (m, 0.76H), 2.96-2.91 (m, 0.76H), 2.86-2.80 (m, 0.24H), 2.77-2.67 (m, 1H), 1.33 (s, 2.16H), 1.15 (s, 6.84H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 161.9 (d, *J* = 243.4 Hz), 161.8 (d, *J* = 241.5 Hz), 154.6, 154.5, 137.3, 134.8, 132.1, 132.1, 129.3, 128.4, 128.3, 127.5, 127.4, 126.9, 126.8, 126.2, 126.1, 124.1, 123.8, 115.4, 115.1, 79.6, 79.5, 55.3, 54.7, 38.9, 36.6, 36.4, 35.6, 28.9, 28.7, 28.5, 28.1 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -118.5, -118.6 ppm; HRMS (ESI-Orbitrap) *m/z*: [M + Na]$^+$ Calcd for $C_{21}H_{24}FNO_2Na^+$: 364.1683, found: 364.1684.

***tert*-Butyl 1-(3-(trifluoromethyl)benzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9d.** Whitefoam (145 mg, 74%); IR (film): ν_{max} 2976, 2931, 1692, 1452, 1418, 1366, 1330, 1164, 1123, 1074, 960, 749, 702cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.53-7.25 (m, 4H), 7.21-7.08 (m, 3H), 7.07-7.00 (m, 0.64H), 6.93-6.86 (m, 0.36H), 5.37 (dd, *J* = 7.2, 6.4 Hz, 0.36H), 5.21 (dd, *J* = 8.4, 5.6 Hz, 0.64H), 4.28-4.12 (m, 0.64H), 3.85-3.72 (m, 0.36H), 3.38-3.27 (m, 1H), 3.15-3.03 (m, 2H), 3.02-2.77 (m, 1H), 2.76-2.69 (m, 0.64H), 2.64-2.55 (m, 0.36H), 1.39 (s, 3.24H), 1.20 (s, 5.76H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.8, 154.4, 139.7, 139.3, 136.7, 134.9, 134.7, 133.2, 130.8 (q, *J* = 31.6 Hz), 129.3, 128.9, 128.7, 128.6, 127.5, 127.3, 127.1, 127.0, 126.5, 126.4, 126.2, 125.7, 123.4 (d, *J* = 3.3 Hz), 123.2 (d, *J* = 3.0 Hz), 79.9, 56.8, 55.6, 43.0, 42.6, 39.5, 37.2, 28.7, 28.6, 28.4, 28.1 ppm; ^{19}F NMR (376 MHz, $CDCl_3$) δ -62.5 ppm; HRMS (ESI-Orbitrap) *m/z*: [M + Na]$^+$ Calcd for $C_{22}H_{24}F_3NO_2Na^+$: 414.1651, found: 414.1650.

***tert*-Butyl 1-(4-cyanobenzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9e.** Whitefoam (157 mg, 90%); IR (film): ν_{max}2974, 2929, 1688, 1418, 1365, 1245, 1163, 1120, 750cm^{-1};^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 8.02-7.98 (m, 0.58H), 7.87-7.83 (m, 0.42H), 7.58-7.57 (m, 0.58H), 7.53-7.52 (m, 0.42H), 7.25-7.18 (m, 3H), 7.16-7.12 (m, 1H), 7.08-7.01 (m, 0.58H), 6.93-6.87 (m, 0.42H), 5.37 (dd, *J* = 7.2, 6.4 Hz, 0.42H), 5.21 (dd, *J* = 8.4, 5.6 Hz, 0.58H), 4.25-4.11 (m, 0.58H), 3.85-3.74 (m, 0.42H), 3.41-3.33 (m, 0.42H), 3.31-3.22 (m, 0.58H), 3.16-3.06 (m, 2H), 2.97-2.88 (m, 0.58H), 2.86-2.78 (m, 0.42H), 2.74-2.62 (m, 1H), 1.39 (s, 3.78H), 1.23 (s, 5.22H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 190.7, 154.8, 154.3, 144.4, 144.1, 138.9, 136.7, 136.3, 134.9, 134.6, 133.0, 132.2, 131.9, 130.6, 130.0, 129.4, 128.8, 127.4, 127.2, 126.2, 119.2, 119.0, 117.8, 117.7, 110.4, 110.3, 80.1, 80.0, 56.5, 55.6, 43.2, 43.1, 39.5, 37.3, 28.7, 28.5, 28.5, 28.3 ppm; HRMS (ESI-Orbitrap) *m/z*: [M + Na]$^+$ Calcd for $C_{22}H_{24}N_2O_2Na^+$: 371.1730, found: 371.1726.

***tert*-Butyl 1-(naphthalen-1-ylmethyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9f.** Whitefoam (129 mg, 69%); IR (film): ν_{max} 2975, 2931, 1688, 1453, 1420, 1240, 1166, 1120, 778, 757 cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 9.29-8.23 (m, 1H), 7.89-7.86 (m, 0.8H), 7.82-7.80 (m, 0.2H), 7.76-7.74 (m, 0.8H), 7.71-7.67 (m, 0.2H), 7.61-7.52 (m, 1H), 7.52-7.47 (m, 1H), 7.40-7.32 (m, 1H), 7.25-7.16 (m, 3.8H), 7.14-7.12 (m, 0.8H), 6.95-6.90 (m, 0.2H), 6.60 (d, *J* =7.6Hz, 0.2H), 5.56-5.51 (m, 0.2H), 5.51-5.47 (m, 0.8H), 3.72-3.66 (m, 0.2H), 3.64-3.57 (m, 1H), 3.48-3.42 (m, 1H), 3.40-3.30 (m, 1H), 3.02-2.93 (m, 0.8H), 2.83-2.76 (m, 1H), 2.67-2.59 (m, 0.2H), 1.41 (s, 1.8H), 0.76 (s, 7.2H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.9, 154.4, 137.5, 136.9, 134.7, 134.6, 134.0, 133.8, 132.8, 132.5, 129.4, 129.1, 128.6, 128.2, 128.2, 128.1, 127.9, 127.4, 127.2, 127.1, 126.8, 126.7, 126.2, 126.1, 125.8, 125.7, 125.6, 125.5, 125.2, 124.3, 123.6, 79.5, 79.3, 55.5, 55.3, 40.1, 40.0, 39.6, 36.5, 28.7, 28.5, 27.6 ppm; HRMS (ESI-Orbitrap) *m/z*: [M + Na]$^+$ Calcd for $C_{25}H_{27}NO_2Na^+$: 396.1934, found: 396.1932.

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***tert*-Butyl 1-(3-bromobenzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9g.** Whitefoam (143 mg, 71%); IR (film): ν_{max}2973, 2928, 1690, 1418, 1365, 1243, 1162, 1118, 762cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.38-7.35 (m, 0.68H), 7.35-7.29 (m, 1H), 7.21-7.14 (m, 3H), 7.14-7.05 (m, 2H), 7.05-7.01 (m, 1H), 6.92-6.86 (m, 0.32H), 5.34 (dd, *J* = 7.2, 6.4 Hz, 0.32H), 5.21 (dd, *J* = 8.4, 6.0 Hz, 0.68H), 4.29-4.17 (m, 0.68H), 3.85-3.75 (m, 0.32H), 3.39-3.32 (m, 0.32H), 3.31-3.22 (m, 0.68H), 3.05-2.97 (m, 2H), 2.96-2.87 (m, 0.68H), 2.86-2.77 (m, 0.32H), 2.76-2.68 (m, 0.68H), 2.67-2.59 (m, 0.32H), 1.41 (s, 2.88H), 1.23(s, 6.12H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.8, 154.4, 141.2, 140.8, 136.9, 136.8, 134.8, 134.7, 132.8, 132.7, 130.0, 129.7, 129.6, 129.5, 129.3, 128.7, 128.5, 127.6, 127.3, 127.0, 126.9, 126.1, 122.5, 122.2, 79.9, 79.8, 56.7, 55.6, 42.8, 42.5, 39.5, 37.0, 28.8, 28.6, 28.5, 28.2 ppm; HRMS (ESI-Orbitrap) *m/z*: [M + Na]$^+$ Calcd for $C_{21}H_{24}BrNO_2Na^+$: 424.0883, 426.0862, found: 424.0881, 426.0861.

***tert*-Butyl 1-(4-fluorobenzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9h.** Whitefoam (126 mg, 74%); IR (film): ν_{max}2974, 2929, 1690, 1509, 1417, 1365, 1222, 1163, 1119, 958, 749cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.21-7.13 (m, 2H), 7.13-7.08 (m, 1H), 7.08-6.98 (m, 3H), 6.97-6.87 (m, 2H), 5.32 (dd, *J* = 7.2, 6.4 Hz, 0.35H), 5.17(dd, *J* = 8.0, 6.0 Hz, 0.65H), 4.25-4.11 (m, 0.65H), 3.84-3.73 (m, 0.35H), 3.39-3.31 (m, 0.35H), 3.29-3.20 (m, 0.65H), 3.07-2.97 (m, 2H), 2.96-2.87 (m, 0.65H), 2.84-2.76 (m, 0.35H), 2.72-2.65 (m, 0.65H), 2.64-2.57 (m, 0.35H), 1.41 (s, 3.15H), 1.25 (s, 5.85H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 161.9 (d, *J* = 242.8 Hz), 161.8 (d, *J* = 242.3 Hz), 154.8, 154.6, 136.9, 134.9, 134.7, 134.4 (d, *J* = 5.8 Hz), 134.1 (d, *J* = 3.0 Hz), 131.2, 131.1, 129.2, 128.6, 127.6, 127.3, 126.9, 126.8, 126.1, 115.2 (d, *J* = 21.0 Hz), 114.9 (d, *J* = 21.1 Hz), 79.8, 79.7, 56.9, 55.9, 42.2, 42.0, 39.5, 37.2, 28.8, 28.6, 28.5, 28.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -116.9, -117.1 ppm; HRMS (ESI-Orbitrap) *m/z*: [M + Na]$^+$ Calcd for $C_{21}H_{24}FNO_2Na^+$: 364.1683, found: 364.1681.

***tert*-Butyl 1-(4-(trifluoromethyl)benzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9i.** White solid (141 mg, 72%); m.p. 109-111 ºC. IR (film): ν_{max} 2976, 2931, 1692, 1418, 1366, 1325, 1245, 1165, 1123, 1067, 752cm^{-1}; ^{1}H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.57-7.42 (m, 2H), 7.27-7.24 (m, 1H), 7.23-7.11 (m, 4H), 7.10-7.07 (m, 0.63H), 6.97-6.88 (m, 0.37H), 5.39 (dd, *J* = 10.8, 6.4 Hz, 0.37H), 5.22 (dd, *J* = 8.4, 5.6 Hz, 0.63H), 4.30-4.17 (m, 0.63H), 3.87-3.74 (m, 0.37H), 3.39-3.23 (m, 1H), 3.16-3.04 (m, 2H), 2.98-2.89 (m, 0.63H), 2.86-2.79 (m, 0.37H), 2.76-2.68 (m, 0.63H), 2.67-2.60 (m, 0.37H), 1.37 (s, 3.33H), 1.18 (s, 5.67H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.7, 154.4, 142.9, 142.5, 136.7, 134.9, 134.6, 130.1, 129.5, 129.4, 129.0 (q, *J* = 32.7 Hz), 128.8, 128.8, 128.5,127.5, 127.2, 127.0, 126.9, 126.2, 125.4 (d, *J* = 3.4 Hz), 125.5 (d, *J* = 3.3 Hz) , 80.0, 79.8, 56.7, 55.5, 42.9, 42.7, 39.4, 37.1, 28.8, 28.6, 28.5, 28.2 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -62.3, -62.4

ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + Na]^+$ Calcd for $C_{22}H_{24}F_3NO_2Na^+$: 414.1651, found: 414.1651.

***tert*-Butyl 1-(4-(trifluoromethoxy)benzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9j.** Whitefoam (134 mg, 66%); IR (film): ν_{max}2975, 2930, 1692, 1418, 1366, 1260, 1163, 1120, 754cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.21-7.16 (m, 2H), 7.16-7.13 (m, 3H), 7.13-7.09 (m, 2H), 7.09-7.06 (m, 0.66H), 6.95-6.91 (m, 0.34H), 5.37 (dd, *J* = 7.6, 6.8 Hz, 0.34H), 5.20 (dd, *J* = 8.4, 5.6 Hz, 0.66H), 4.27-4.19 (m, 0.66H), 3.86-3.78 (m, 0.34H), 3.38-3.34 (m, 0.36H), 3.31-3.23 (m, 0.64H), 3.09-3.01 (m, 2H), 2.98-2.91 (m, 0.64H), 2.85-2.78 (m, 0.36H), 2.75-2.67 (m, 0.64H), 2.64-2.59 (m, 0.36H), 1.37 (s, 3.06H), 1.20 (s, 5.94H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.7, 154.5, 148.1, 148.0, 137.6, 137.2, 136.9, 134.9, 134.6, 131.0, 129.3, 128.7, 127.5, 127.3, 127.0, 126.9, 126.2, 121.1, 120.7, 120.6 (q, *J* = 255.1 Hz), 119.4, 79.9, 79.8, 56.8, 55.6, 42.4, 42.2, 39.3, 37.0, 28.8, 28.6, 28.4, 28.2 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -110.6, -111.0 ppm; HRMS (ESI-Orbitrap) *m/z*:$[M + Na]^+$ Calcd for $C_{22}H_{24}F_3NO_3Na^+$: 430.1601, found: 430.1600.

***tert*-Butyl 1-(2,6-difluorobenzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9k.** Whitefoam (140 mg, 78%); IR (film): ν_{max} 2976, 2931, 1694, 1469, 1419, 1366, 1166, 1119, 1029, 781, 747cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.24-7.17 (m, 4H), 7.17-7.09 (m, 1H), 6.90-6.79 (m, 2H), 5.52 (dd, J= 5.6, 4.8 Hz, 0.24H), 5.38 (dd, J= 10.8, 4 Hz, 0.76H), 4.32-4.23 (m, 0.76H), 4.15-3.95 (m, 0.24H), 3.50-3.43 (m, 0.24H), 3.42-3.31 (m, 0.76H), 3.25-3.13 (m, 1H), 3.11-3.01 (m, 1H), 3.00-2.91 (m, 0.76H), 2.90-2.82 (m, 0.24H), 2.79-2.71 (m, 1H), 1.28 (s, 2.16H), 1.15 (s, 6.84H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 162.4 (d, *J* = 245.6 Hz), 162.3 (d, *J* = 245.6 Hz), 161.2, 161.1, 154.5, 154.3, 137.2, 137.1, 134.9, 134.6, 129.4, 128.9, 128.2， 128.1, 128.0, 128.0, 127.5, 127.3, 127.0, 126.9, 126.8, 126.3, 126.2, 114.8 (t, *J* = 20.2 Hz), 114,6 (t, *J* = 20.2 Hz), 111.2, 111.1, 111.0, 110.9, 110.8, 110.7, 79.6, 79.4, 54.5, 53.5, 38.3, 36.4, 29.4, 29.3, 28.9, 28.6, 28.3, 28.0 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -115.5, -115.7 ppm; HRMS (ESI-Orbitrap) *m/z*: $[M + Na]^+$ Calcd for $C_{21}H_{23}F_2NO_2Na^+$: 382.1589, found: 382.1594.

***tert*-Butyl 1-(3,5-difluorobenzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9l.** Whitefoam (115 mg, 64%); IR (film): ν_{max}2975, 2931, 1691, 1624, 1595, 1417, 1366, 1330, 1163, 1117, 985, 850, 747cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.21-7.11 (m, 3H), 7.07-6.98 (m, 0.63H), 6.96-6.90 (m, 0.37H), 6.73-6.58 (m, 3H), 5.34 (dd, *J* = 7.2, 6.4 Hz, 0.37H), 5.22 (dd, *J* = 8.4, 5.6 Hz, 0.63H), 4.24-4.15 (m, 0.63H), 3.89-3.77 (m, 0.37H), 3.37-3.32 (m, 0.37H), 3.29-3.18 (m, 0.63H), 3.06-2.67 (m, 4H), 1.41 (s, 3.33H), 1.28 (s, 5.67H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 163.1 (d, *J* = 246.3 Hz), 163.0 (d, *J* = 245.0 Hz), 154.8, 154.4, 142.6, 142.3, 136.5, 134.8, 134.6, 129.4, 128.8, 127.4, 127.2, 127.1, 126.2, 112.7, 112.6, 112.5, 112.4, 102.0 (t, *J* = 25.5 Hz), 101.9 (t, *J* = 25.1 Hz), 80.0, 56.4, 55.5, 42.9, 42.6, 39.4, 37.2, 28.8, 28.5, 28.5, 28.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, mixture of rotamers) δ -110.6, -111.0 ppm; HRMS (ESI-Orbitrap) *m/z*:$[M + Na]^+$ Calcd for $C_{21}H_{23}F_2NO_2Na^+$: 382.1589, found: 382.1584.

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***tert*-Butyl 1-(4-(methylsulfonyl)benzyl)-3,4-dihydroisoquinoline-2(1H)-carboxylate 9m.** Whitefoam (151 mg, 75%); IR (film): ν_{max} 2974, 2928, 1687, 1418, 1303, 1150, 1091, 957, 759cm^{-1}; ^1H NMR (400 MHz, $CDCl_3$, mixture of rotamers) δ 7.89-7.85 (m, 1H), 7.84-7.77 (m, 1H), 7.36-7.33 (m, 1H), 7.33-7.27 (m, 1H), 7.21-7.11 (m, 3H), 7.07-6.92 (m, 1H), 5.40 (dd, *J* = 7.2, 6.4Hz, 0.44H), 5.25 (dd, *J* = 6.4, 5.6 Hz, 0.56H), 4.25-4.16 (m, 0.56H), 3.87-3.78 (m, 0.44H), 3.40-3.25 (m, 1H), 3.18-3.11 (m, 2H), 3.04-3.01 (m, 3H), 2.96-2.79 (m, 1H), 2.76-2.63 (m, 1H), 1.38 (s, 3.96H), 1.22 (s, 5.04H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, mixture of rotamers) δ 154.7, 154.3, 145.3, 145.1, 138.9, 138.5, 136.3, 134.8, 134.6, 130.7, 129.3, 128.8, 127.5, 127.3, 127.2, 127.1, 127.0, 126.2, 79.9, 79.8, 56.5, 55.5, 44.6, 43.0, 42.8, 39.4, 37.2, 28.7, 28.5, 28.4, 28.2 ppm; HRMS (ESI-Orbitrap) *m/z*:$[M + Na]^+$ Calcd for $C_{22}H_{27}NO_4SNa^+$: 424.1553, found: 424.1551.

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